# Purchase Instructions for Xiaozhi AI ESP32S3 Development Board + Breadboard DIY Module

Regarding the DIY replication of the breadboard solution for Xiaozhi AI Chatbot, instructions for purchasing components/modules of relevant hardware terminals:

This project is mainly open source, but we will select some links that group members often purchase and include them in the document for everyone's reference. For more, please search on Taobao or other shopping platforms by yourself.

The only thing that requires special attention is to make sure to use the development board with official modules!

Additionally, the signal of the development board chip module of Guoyun on the market is stable, and QQ group members have reported fewer signal issues, so it can be used.

Please try to choose development boards that use official modules. The ESP32-S3-WROOM-1 official module is shown below:

**1)**Search for loose parts by module name, for example, search for the development board with "**ESP32 S3 Development Board**", search for the microphone with "**INMP441**", search for the audio amplifier with "**MAX98357**". Generally, you can purchase all the modules on the list from the same store (except for the ML307R-4G module).

**2)** For the combined kit, please search for the keywords " **Xiaozhi AI ESP32** " or " **Xiaozhi AI ESP32S3** " etc., and feel free to search for others on your own.

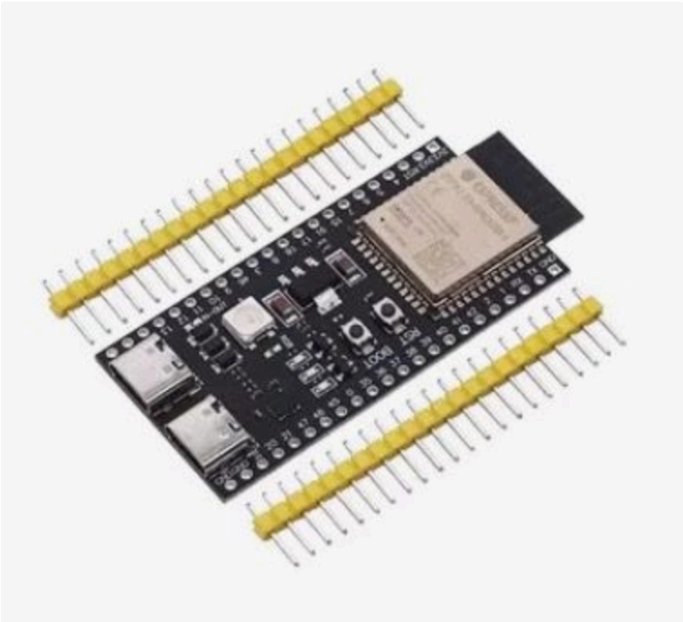
**Note:** Due to differences in merchants' products, please try to choose stores with high credibility, high sales volume, and good reviews for purchases. For purchase consultations and after-sales quality issues, please negotiate with the merchant on your own, and this project is not responsible.

# 1. ESP32-S3-DevKitC-1 Development Board and Component Purchase (Loose Parts)

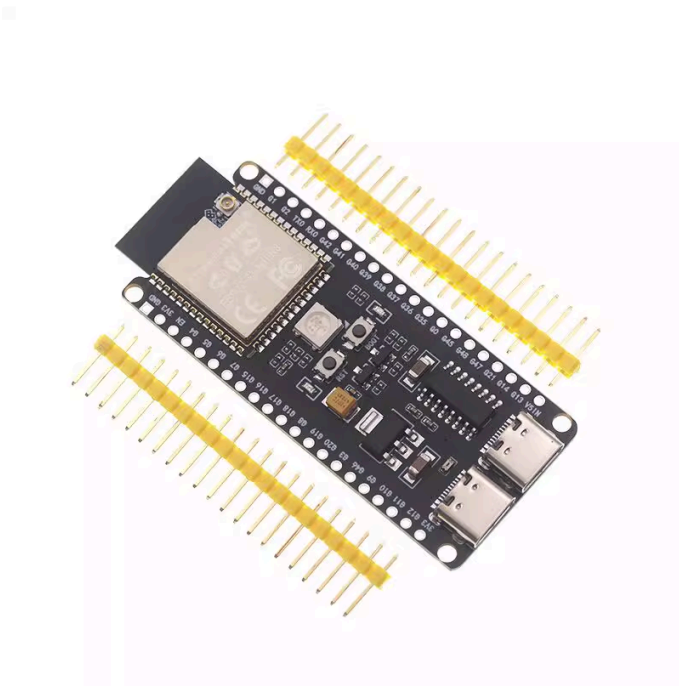
## 2.1 ESP32-S3-DevKitC-1 Development Board

The hardware solution of Xiaozhi AI Chatbot mainly uses **the Espressif ESP32-S3-DevKitC-1 development board** (equipped with **the ESP32-S3-WROOM-1/1U N16R8** chip module).

Currently, there are mainly the Espressif Standard Edition (Version 1) with 44 IO pins and the GOOUUU Guoyun Edition (Version 2) with 42 IO pins on the market, both of which are equipped with**ESP32-S3-WROOM-1/1U N16R8**chip modules, and are supported by Xiaozhi AI Chatbot. The main difference lies in the slightly different positions of the IO interfaces, as shown in the following illustration:



Espressif Standard Edition with 44 IO pins



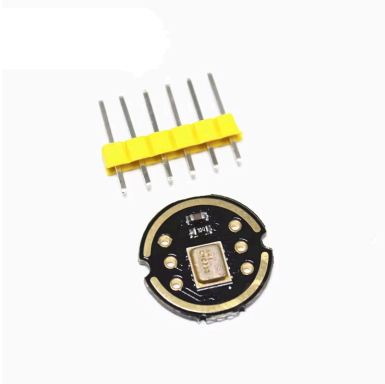
GOOUUU Fruit Cloud Edition with 42 IO pins

## 2.2 MEMS Digital Microphone (INMP441)

**1)** It is recommended to purchase accessories with soldered pin headers, which are convenient for plugging into breadboards;

**2)** When purchasing a microphone, search for **"INMP441"**;

**3)** The price is generally between 5.5 and 9 yuan, and it seems to have increased recently.



## 2.3 Digital Power Amplifier (MAX98357A)

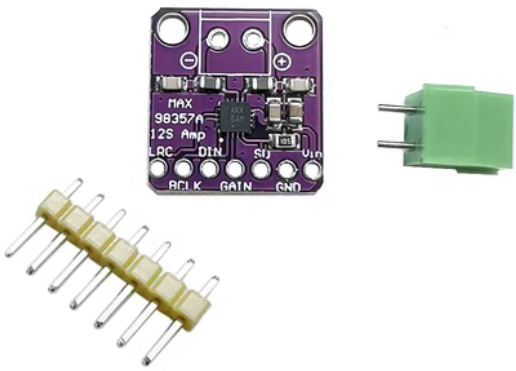
**1)** It is recommended to purchase accessories with soldered pin headers, which are convenient for plugging into breadboards;

**2)** When purchasing a digital audio power amplifier, please search for "**MAX98357**" or "**MAX98357A**" etc.;

**3)** The price is generally between 3 and 6 yuan;

**4)** Model: MAX98357A , MAX98357B and MAX98357 are all supported and are compatible.

The BGA package of MAX98357 has low cost, but the sound quality may be slightly inferior, and some group members have reported some issues. **It is not recommended to purchase the MAX98357 audio amplifier with BGA package.**




MAX98357A Reference Diagram

## 2.4 Cavity Speaker

**1)** Recommended cavity speakers are (8Ω 3W) / (4Ω 3W); speakers exceeding this power cannot fully perform their functions.

**2)** Search for **Cavity Speaker 3W** ;

**3)** Prices generally range from 2 yuan to 10 yuan, with more expensive ones possibly having better effects. Please choose according to your own preference;

**4)**Speaker terminals, please follow the wiring of the digital power amplifier you purchased. MAX98357 does not require terminal blocks for connection and is fixed with screw holes.

If a more powerful speaker, or stereo dual speakers, etc., are used, additional power supply and audio module support will be required, which users should consider on their own.

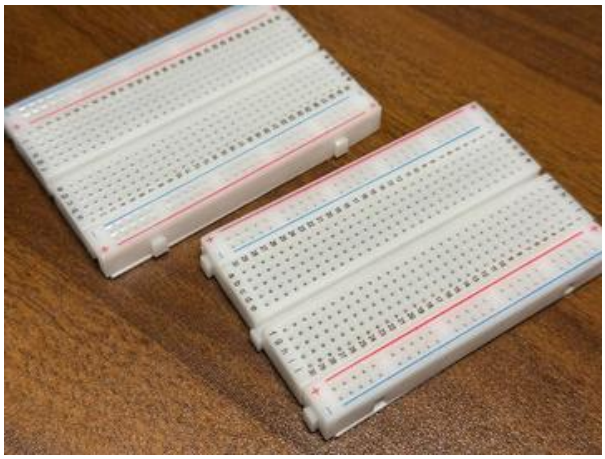
Terminals can be self-selected or omitted

## 2.5 Breadboard (2 pieces, combinable into 400 holes, optional, recommended)

**1)** It is recommended to purchase 2 400-hole breadboards. Without breadboards, as more and more electronic components are connected to the development board, it will easily become a complete mess. If you use DuPont wires or soldering for assembly, you do not need to buy breadboards.

**2)** Purchase Search for 400-hole Breadboard

**3)** Prices range from 1.5 to 5 yuan, and it is recommended to buy a breadboard of better quality.



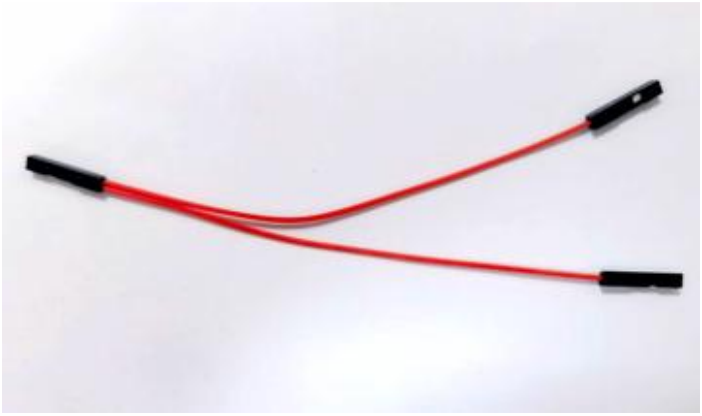
## 2.6 Conductors (1 box of jumpers, several DuPont wires)

**1)** It is recommended to directly purchase a 1-box set, which includes jumpers of various specifications.

**2)** Purchase Search **Breadboard Jumper Wires**

**3)**Reference price: 5-7 yuan

- **Note:** If you use DuPont wires for connection (without using a breadboard), please add a few 1-to-2 branching wires (3 bus wires) when purchasing DuPont wires for short-circuiting the microphone and digital power amplifier to address the issue of insufficient pin headers for the 3V3 power supply and GND ground on the development board.



DuPont Wire 1-to-2 Bus Bar

## 2.7 OLED Display (IIC Interface, Recommended, Optional)

**1)** When purchasing, it is necessary to confirm that it uses the SSD1306 driver. It is recommended to choose a screen with the pin starting with GND (consistent with the wiring tutorial). The default size is 0.91 inches, and 0.96 inches or 1.3 inches OLED screens driven by SSD1306 are also supported, but the resolution needs to be adjusted (compiled in the Development Environment).

**2)** Search **SSD 1306 OLED Screen**

**3)**Reference price is between 5 and 7 yuan

## 2.8 **ML307R - 4G Cat1 Communication Module (Optional)**

**Note: When purchasing the CMCC ML307R 4G Cat1 communication module,please be sure to select the AT firmware version!!! Other versions are not supported for the time being!!!**

**1)**Please select**the AT firmware version of the ML307R 4G Cat1 communication module;**

**2) Both the ML307R-DL and ML307R-DC versions are available.** Regular users can simply purchase the DL version, while developers can choose the DC version (which supports openCPU, etc.; please refer to the details on your own).

**3)** It is recommended to purchase a pluggable card module with a SIM card holder to facilitate the replacement of 4G data cards. **Surface-mount cards are soldered onto the PCB board, so try not to choose them.** You can purchase additional data cards or use your own mobile phone card (with data traffic).

**4)** It is recommended to choose an antenna when purchasing to enhance 4G mobile signals.

**Reference for purchase link:**

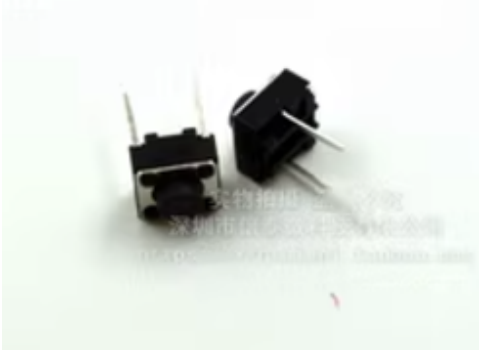
Xiaolanqing IoT Technology:https://item.taobao.com/item.htm?from=cart&id=797466121802&skuId=5434497184727

Du Yun IoT: https://detail.tmall.com/item.htm?_u=ko1bsr22a1&id=737949327567

## 2.9 Microswitch/Button

Use a 6*6MM micro switch/tactile switch. On the breadboard, stagger the rows and insert it into the breadboard according to the schematic diagram. The buttons are as shown in the following figure:

**Note:** To prevent beginners from short-circuiting pins, Brother Xia recommends using a 2-pin through-hole button instead of a 4-pin one (experienced users can ignore this).



Reference for purchase link:https://item.taobao.com/item.htm?id=42823169259

# 2. ESP32-S3-DevKitC-1 Development Board and Breadboard Kit

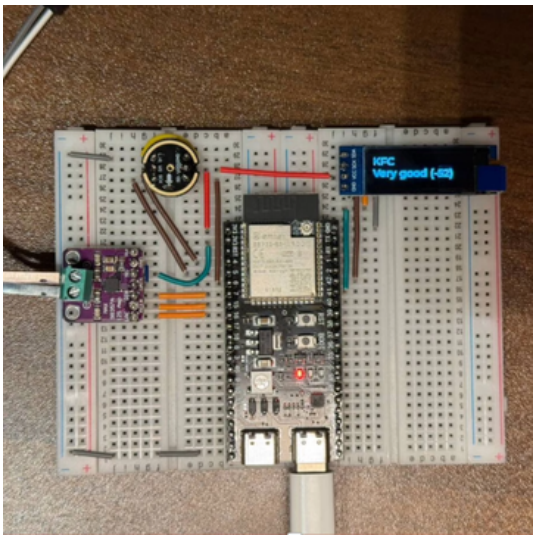
Xiaozhi AI Chatbot Terminal Hardware, **Breadboard DIY Solution** Please purchase the ESP32-S3 development board and breadboard kit, which can save the time spent searching for and comparing components, facilitate users to make a one-time purchase, and may save on shipping costs compared to purchasing individual components. Users can choose to purchase at their own discretion.

Breadboard Xiaozhi AI WIFI Combo Kit (excluding 4G module), plug in wires according to the wiring tutorial (no soldering required), **recommended for beginners: Breadboard (assembled) version;**

Reference link: to be supplemented

You can also search for the Xiaozhi AI Breadboard Kit on shopping platforms and compare to make a selection.

Preview is as shown in the following illustration:




**Note:**New users are advised to purchase integrated kits (wiring-free), and it is recommended to buy Xiaozhi AI official commercial products or third-party development boards mentioned in the documentation, as detailed in Sections 3 and 4. For other development boards, please search and purchase them on Taobao or other platforms on your own.